confocal_lacerate

```r
library(tidyverse)
library(dplyr)
library(stringr)
library(rstatix)
library(patchwork)
library(car)
library(emmeans)
library(multcomp)
library(multcompView)
library(tibble)

rm(list = ls())
#graphics.off()

select <- dplyr::select
filter <- dplyr::filter
mutate <- dplyr::mutate

getwd()
```

```
## [1] "/Users/junbc/Documents/GitHub/heating_lacerates_final/code"
```

```r
setwd("~/Documents/GitHub/heating_lacerates_final")

# Data Processing
raw_confocal_data <- read_csv("data/Lacerate_EdU_Caspase.csv")

confocal_data <- raw_confocal_data %>%
  filter(True_Channel %in% c("EdU", "Caspase")) %>%
  mutate(
    marker = factor(True_Channel, levels = c("EdU", "Caspase")),
    time = factor(Time, levels = c("2dpl", "5dpl", "8dpl", "14dpl")),
    treatment = as.character(Treatment),
    state = case_when(
      str_detect(treatment, regex("Apo", ignore_case = TRUE)) ~ "Apo",
      str_detect(treatment, regex("Inoc", ignore_case = TRUE)) ~ "Inoc",
      str_detect(treatment, regex("Sym", ignore_case = TRUE)) ~ "Sym",
      TRUE ~ NA_character_
    ),
    heat = case_when(
      str_detect(treatment, regex("HS|Heat", ignore_case = TRUE)) ~ "Heat",
      str_detect(treatment, regex("Control", ignore_case = TRUE)) ~ "Control",
      TRUE ~ NA_character_
    ),
    state = factor(state, levels = c("Apo", "Inoc", "Sym")),
```

```
    heat = factor(heat, levels = c("Control", "Heat")),
    percentage = Percentage
  ) %>%
  drop_na(marker, state, heat, time, percentage)

sample_size_table_wide <- confocal_data %>%
  count(marker, state, time, heat) %>%
  pivot_wider(names_from = heat, values_from = n)

print(sample_size_table_wide)
```

```
## # A tibble: 22 x 5
##    marker state time  Control  Heat
##    <fct>  <fct> <fct>   <int> <int>
##  1 EdU    Apo   2dpl        3     6
##  2 EdU    Apo   5dpl        3     5
##  3 EdU    Apo   8dpl        5     4
##  4 EdU    Apo   14dpl       5     4
##  5 EdU    Inoc  5dpl        4     5
##  6 EdU    Inoc  8dpl        4     5
##  7 EdU    Inoc  14dpl       7     6
##  8 EdU    Sym   2dpl        5     6
##  9 EdU    Sym   5dpl       10     8
## 10 EdU    Sym   8dpl        6     8
## # i 12 more rows
```

```
p6_n <- confocal_data %>%
  count(marker, state, time, heat) %>%
  arrange(marker, state, time, heat)

print(p6_n, n = Inf)
```

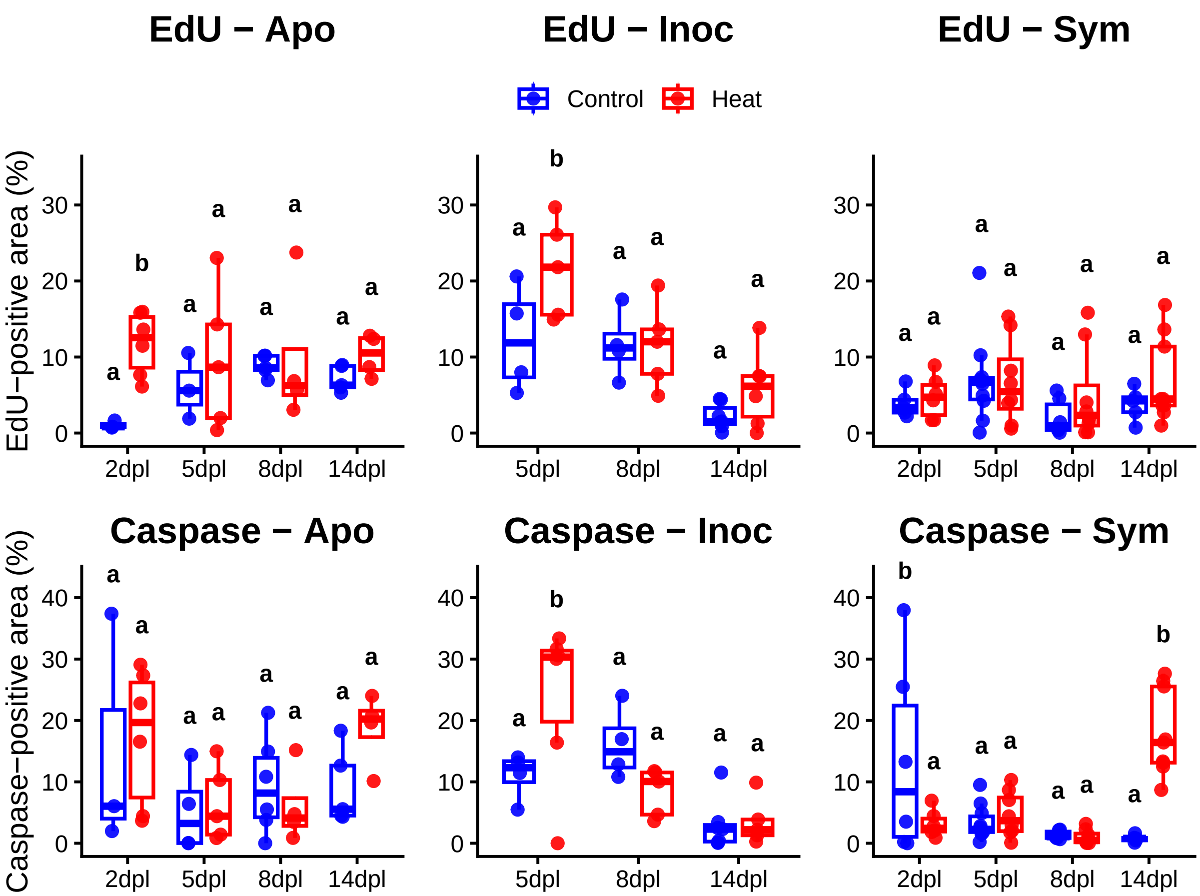
```
## # A tibble: 44 x 5
##    marker   state time  heat        n
##    <fct>    <fct> <fct> <fct>   <int>
##  1 EdU      Apo   2dpl  Control     3
##  2 EdU      Apo   2dpl  Heat        6
##  3 EdU      Apo   5dpl  Control     3
##  4 EdU      Apo   5dpl  Heat        5
##  5 EdU      Apo   8dpl  Control     5
##  6 EdU      Apo   8dpl  Heat        4
##  7 EdU      Apo   14dpl Control     5
##  8 EdU      Apo   14dpl Heat        4
##  9 EdU      Inoc  5dpl  Control     4
## 10 EdU      Inoc  5dpl  Heat        5
## 11 EdU      Inoc  8dpl  Control     4
## 12 EdU      Inoc  8dpl  Heat        5
## 13 EdU      Inoc  14dpl Control     7
## 14 EdU      Inoc  14dpl Heat        6
## 15 EdU      Sym   2dpl  Control     5
## 16 EdU      Sym   2dpl  Heat        6
## 17 EdU      Sym   5dpl  Control    10
```

```
## 18 EdU     Sym   5dpl  Heat        8
## 19 EdU     Sym   8dpl  Control     6
## 20 EdU     Sym   8dpl  Heat        8
## 21 EdU     Sym   14dpl Control     5
## 22 EdU     Sym   14dpl Heat        9
## 23 Caspase Apo   2dpl  Control     3
## 24 Caspase Apo   2dpl  Heat        6
## 25 Caspase Apo   5dpl  Control     4
## 26 Caspase Apo   5dpl  Heat        5
## 27 Caspase Apo   8dpl  Control     6
## 28 Caspase Apo   8dpl  Heat        4
## 29 Caspase Apo   14dpl Control     5
## 30 Caspase Apo   14dpl Heat        4
## 31 Caspase Inoc  5dpl  Control     4
## 32 Caspase Inoc  5dpl  Heat        6
## 33 Caspase Inoc  8dpl  Control     4
## 34 Caspase Inoc  8dpl  Heat        5
## 35 Caspase Inoc  14dpl Control     7
## 36 Caspase Inoc  14dpl Heat        5
## 37 Caspase Sym   2dpl  Control     6
## 38 Caspase Sym   2dpl  Heat        6
## 39 Caspase Sym   5dpl  Control    10
## 40 Caspase Sym   5dpl  Heat        8
## 41 Caspase Sym   8dpl  Control     6
## 42 Caspase Sym   8dpl  Heat        8
## 43 Caspase Sym   14dpl Control     5
## 44 Caspase Sym   14dpl Heat        9
```

```
ymax_edu <- confocal_data %>%
  filter(marker == "EdU") %>%
  summarise(max_val = max(percentage, na.rm = TRUE)) %>%
  pull(max_val)

ymax_cas <- confocal_data %>%
  filter(marker == "Caspase") %>%
  summarise(max_val = max(percentage, na.rm = TRUE)) %>%
  pull(max_val)

ymax_edu <- ymax_edu + 5
ymax_cas <- ymax_cas + 5

make_panel <- function(data_sub, panel_title, ylab = NULL, show_legend = FALSE, y_limit = NULL) {

  fit <- lm(percentage ~ heat * time, data = data_sub)

  emm <- emmeans(fit, ~ heat | time)

  letters_df <- cld(emm, adjust = "tukey", Letters = letters) %>%
    as.data.frame() %>%
    mutate(.group = str_trim(.group))

  y_pos <- data_sub %>%
    group_by(time, heat) %>%
    summarise(y = max(percentage, na.rm = TRUE), .groups = "drop")
```

```r
  letters_df <- left_join(letters_df, y_pos, by = c("time", "heat")) %>%
    mutate(y = y + 6.5)

  if (is.null(y_limit)) {
    y_max <- max(c(data_sub$percentage, letters_df$y), na.rm = TRUE) + 2
  } else {
    y_max <- y_limit
  }

  ggplot(data_sub, aes(x = time, y = percentage, color = heat)) +
    geom_jitter(
      position = position_jitterdodge(jitter.width = 0.12, dodge.width = 0.75),
      size = 2.2,
      alpha = 0.9
    ) +
    geom_boxplot(
      aes(group = interaction(time, heat)),
      outlier.shape = NA,
      width = 0.6,
      fill = NA,
      linewidth = 0.8,
      position = position_dodge(width = 0.75)
    ) +
    geom_text(
      data = letters_df,
      aes(x = time, y = y, label = .group, group = heat),
      position = position_dodge(width = 0.75),
      inherit.aes = FALSE,
      fontface = "bold",
      size = 4,
      color = "black"
    ) +
    scale_color_manual(values = c("Control" = "blue", "Heat" = "red")) +
    coord_cartesian(ylim = c(0, y_max), clip = "off") +
    labs(title = panel_title, x = NULL, y = ylab, color = NULL) +
    theme_classic(base_size = 14) +
    theme(
      plot.title = element_text(face = "bold", hjust = 0.5),
      legend.position = if (show_legend) "top" else "none",
      plot.margin = margin(8, 8, 8, 8)
    )
}

# Figure

# ---- EdU (shared scale)
p_edu_apo <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Apo"),
  "EdU - Apo",
  "EdU-positive area (%)",
  y_limit = ymax_edu
)
```

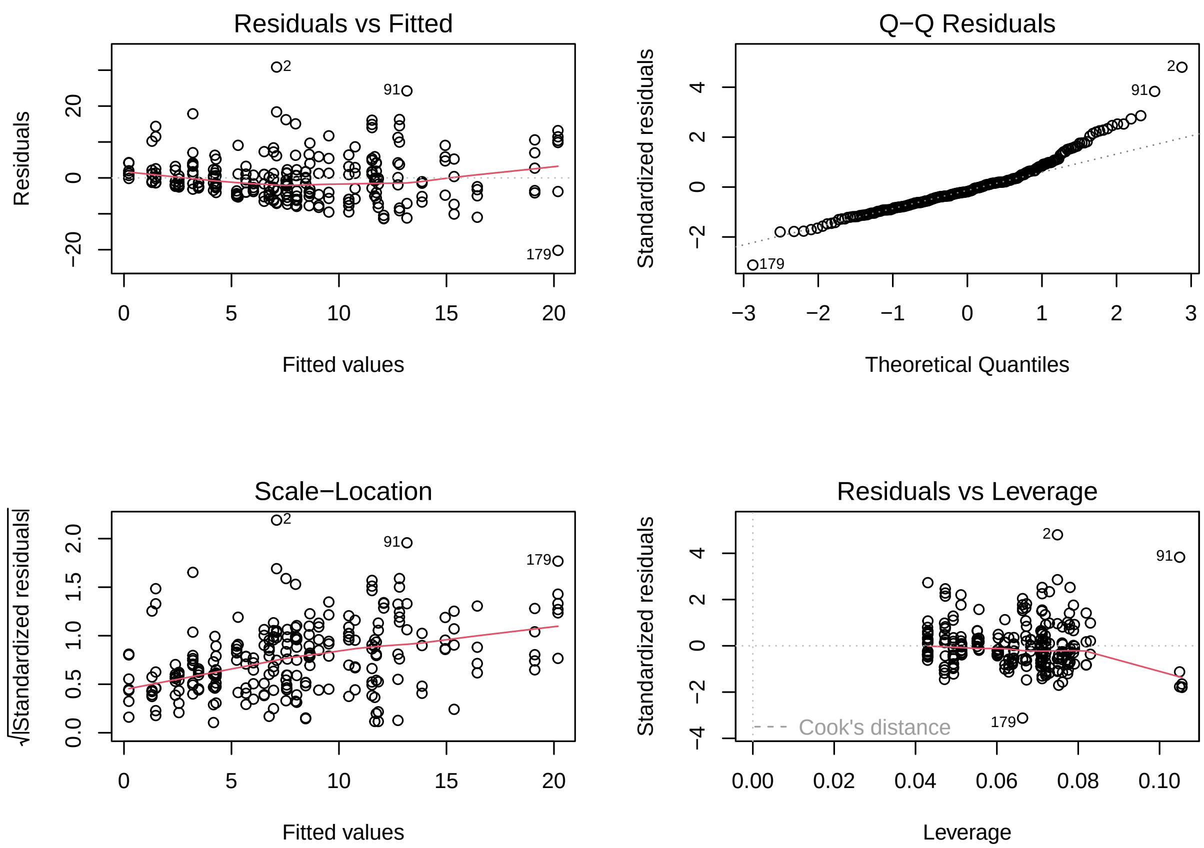
```r
p_edu_inoc <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Inoc"),
  "EdU - Inoc",
  y_limit = ymax_edu
)

p_edu_sym <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Sym"),
  "EdU - Sym",
  show_legend = TRUE,
  y_limit = ymax_edu
)

# ---- Caspase (shared scale)
p_cas_apo <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Apo"),
  "Caspase - Apo",
  "Caspase-positive area (%)",
  y_limit = ymax_cas
)

p_cas_inoc <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Inoc"),
  "Caspase - Inoc",
  y_limit = ymax_cas
)

p_cas_sym <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Sym"),
  "Caspase - Sym",
  show_legend = TRUE,
  y_limit = ymax_cas
)
p_edu3 <- (p_edu_apo | p_edu_inoc | p_edu_sym) +
  plot_layout(guides = "collect") &
  theme(legend.position = "top")

p_cas3 <- (p_cas_apo | p_cas_inoc | p_cas_sym) +
  plot_layout(guides = "collect") &
  theme(legend.position = "top")

p6 <- (p_edu_apo | p_edu_inoc | p_edu_sym) /
  (p_cas_apo | p_cas_inoc | p_cas_sym) +
  plot_layout(guides = "collect") &
  theme(legend.position = "top")

print(p6)
```



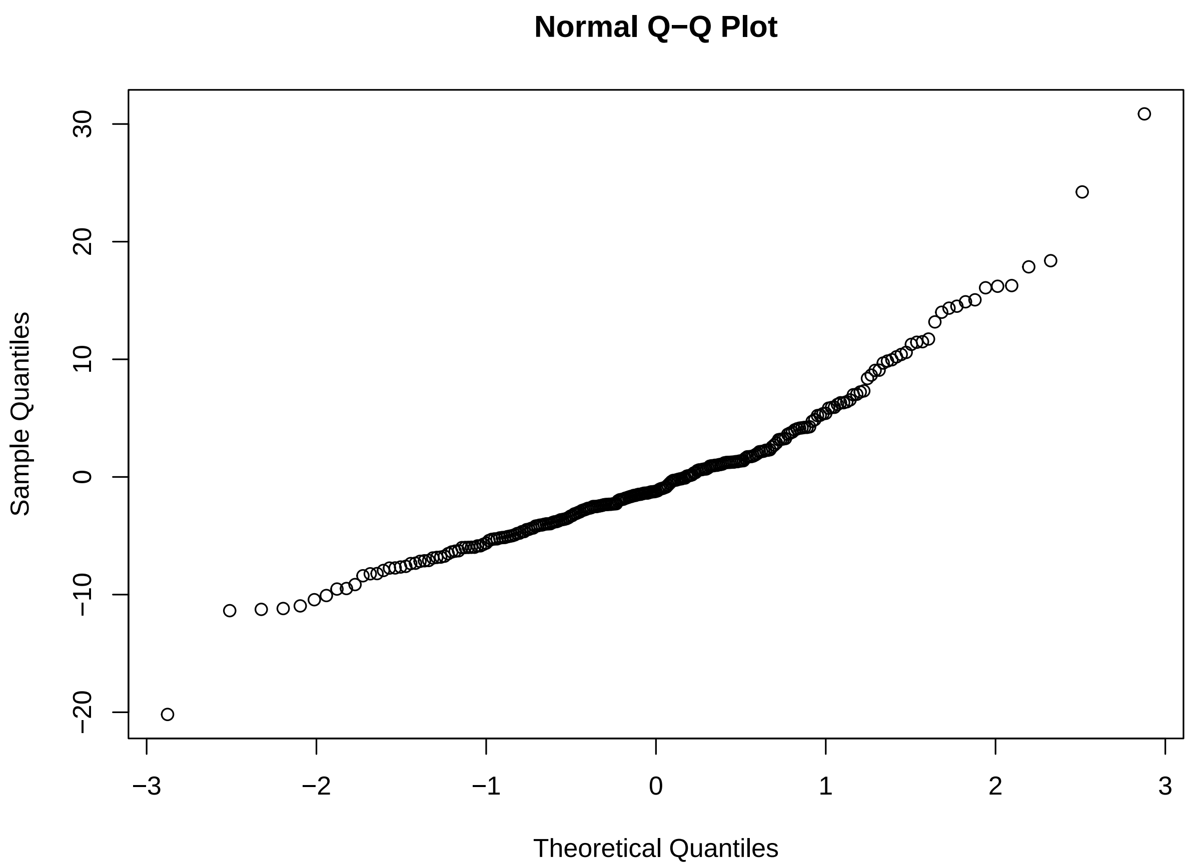

```r
ggsave(
  filename = "Fig6_confocal.png",
  plot = p6,
  path = "figs",
  device = "png",
  width = 11,
  height = 8,
  units = "in",
  dpi = 600,
  bg = "white"
)

ggsave(
  filename = "Fig6_confocal.pdf",
  plot = p6,
  path = "figs",
  device = pdf,
  width = 11,
  height = 8,
  units = "in",
  bg = "white"
)
```

```r
# ---- EdU (shared scale)
p_edu_apo <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Apo"),
  "EdU - Apo",
  "EdU-positive area (%)",
  y_limit = ymax_edu
)

p_edu_inoc <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Inoc"),
  "EdU - Inoc",
  y_limit = ymax_edu
)

p_edu_sym <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Sym"),
  "EdU - Sym",
  show_legend = TRUE,
  y_limit = ymax_edu
)

# ---- Caspase (shared scale)
p_cas_apo <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Apo"),
  "Caspase - Apo",
  "Caspase-positive area (%)",
  y_limit = ymax_cas
)

p_cas_inoc <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Inoc"),
  "Caspase - Inoc",
  y_limit = ymax_cas
)

p_cas_sym <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Sym"),
  "Caspase - Sym",
  show_legend = TRUE,
  y_limit = ymax_cas
)


#Stats

fit_global <- lm(
  percentage ~ marker + state + heat * time + state * time,
  data = confocal_data
)

## Global model checks

par(mfrow = c(2, 2))
```

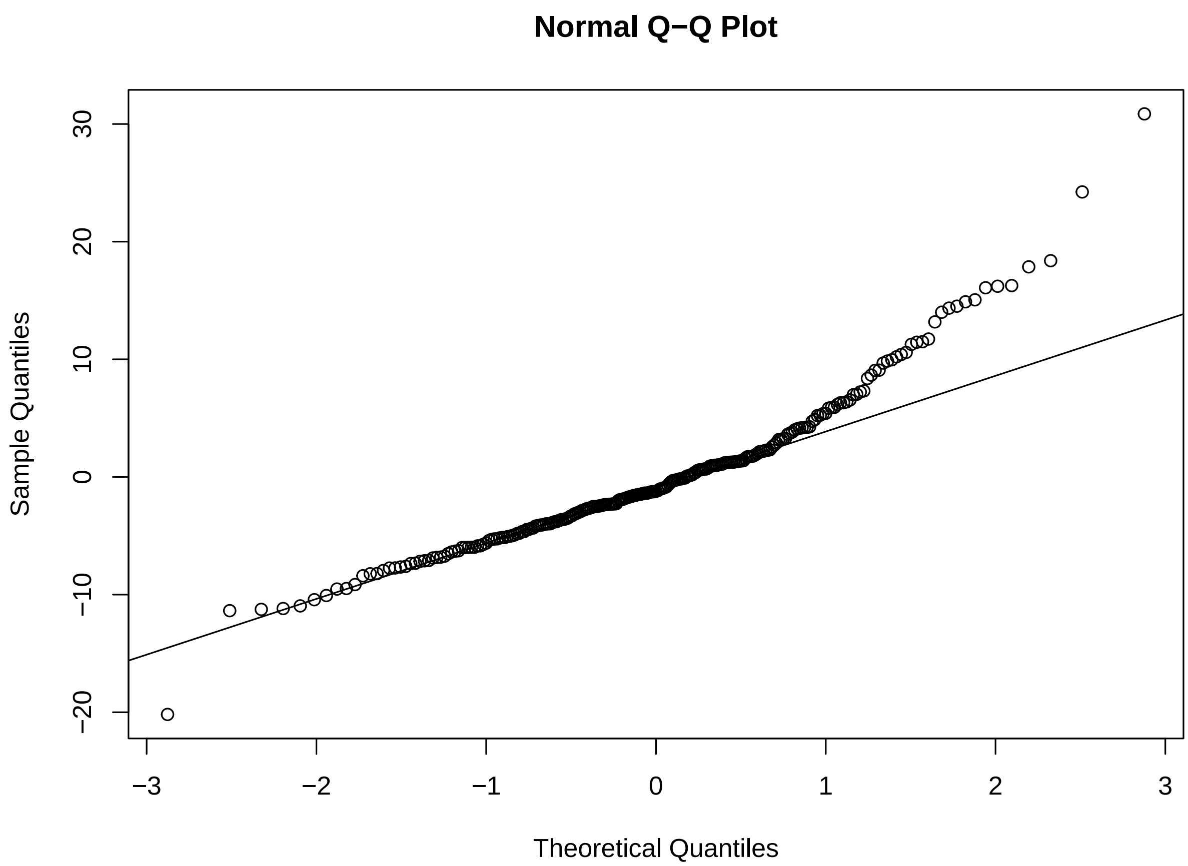
```r
plot(fit_global)
```




```r
par(mfrow = c(1, 1))

qqnorm(residuals(fit_global))
```




```r
qqline(residuals(fit_global))
```




```r
shapiro.test(residuals(fit_global))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(fit_global)
## W = 0.93424, p-value = 4.264e-09
```

```r
leveneTest(
  percentage ~ marker * state * heat * time,
  data = confocal_data
)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value   Pr(>F)
## group  43  1.9537 0.001075 **
##       205
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
## Type II ANOVA

anova_global <- car::Anova(fit_global, type = 2)

anova_global_table <- as.data.frame(anova_global) %>%
  rownames_to_column("term") %>%
  mutate(
    across(where(is.numeric), ~ signif(.x, 3))
  )
anova_global_table
```

```
##          term  Sum Sq  Df F value   Pr(>F)
## 1      marker    72.3   1    1.62 2.05e-01
## 2       state  1450.0   2   16.20 2.61e-07
## 3        heat   400.0   1    8.95 3.07e-03
## 4        time   385.0   3    2.87 3.71e-02
## 5   heat:time   537.0   3    4.00 8.33e-03
## 6  state:time  2460.0   5   11.00 1.55e-09
## 7   Residuals 10400.0 233      NA       NA
```

```r
edu_apo <- confocal_data %>%
  filter(marker == "EdU", state == "Apo") %>%
  droplevels()

fit_edu_apo <- lm(percentage ~ heat * time, data = edu_apo)
anova_edu_apo <- car::Anova(fit_edu_apo, type = 2)
emm_edu_apo <- emmeans(fit_edu_apo, ~ heat | time)
tukey_edu_apo <- pairs(emm_edu_apo, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "EdU", state = "Apo")
letters_edu_apo <- cld(emm_edu_apo, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "EdU", state = "Apo")

edu_inoc <- confocal_data %>%
  filter(marker == "EdU", state == "Inoc") %>%
  droplevels()

fit_edu_inoc <- lm(percentage ~ heat * time, data = edu_inoc)
anova_edu_inoc <- car::Anova(fit_edu_inoc, type = 2)
emm_edu_inoc <- emmeans(fit_edu_inoc, ~ heat | time)
tukey_edu_inoc <- pairs(emm_edu_inoc, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "EdU", state = "Inoc")
letters_edu_inoc <- cld(emm_edu_inoc, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "EdU", state = "Inoc")

edu_sym <- confocal_data %>%
  filter(marker == "EdU", state == "Sym") %>%
  droplevels()

fit_edu_sym <- lm(percentage ~ heat * time, data = edu_sym)
```

```r
anova_edu_sym <- car::Anova(fit_edu_sym, type = 2)
emm_edu_sym <- emmeans(fit_edu_sym, ~ heat | time)
tukey_edu_sym <- pairs(emm_edu_sym, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "EdU", state = "Sym")
letters_edu_sym <- cld(emm_edu_sym, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "EdU", state = "Sym")

cas_apo <- confocal_data %>%
  filter(marker == "Caspase", state == "Apo") %>%
  droplevels()

fit_cas_apo <- lm(percentage ~ heat * time, data = cas_apo)
anova_cas_apo <- car::Anova(fit_cas_apo, type = 2)
emm_cas_apo <- emmeans(fit_cas_apo, ~ heat | time)
tukey_cas_apo <- pairs(emm_cas_apo, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "Caspase", state = "Apo")
letters_cas_apo <- cld(emm_cas_apo, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "Caspase", state = "Apo")

cas_inoc <- confocal_data %>%
  filter(marker == "Caspase", state == "Inoc") %>%
  droplevels()

fit_cas_inoc <- lm(percentage ~ heat * time, data = cas_inoc)
anova_cas_inoc <- car::Anova(fit_cas_inoc, type = 2)
emm_cas_inoc <- emmeans(fit_cas_inoc, ~ heat | time)
tukey_cas_inoc <- pairs(emm_cas_inoc, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "Caspase", state = "Inoc")
letters_cas_inoc <- cld(emm_cas_inoc, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "Caspase", state = "Inoc")

cas_sym <- confocal_data %>%
  filter(marker == "Caspase", state == "Sym") %>%
  droplevels()

fit_cas_sym <- lm(percentage ~ heat * time, data = cas_sym)
anova_cas_sym <- car::Anova(fit_cas_sym, type = 2)
emm_cas_sym <- emmeans(fit_cas_sym, ~ heat | time)
tukey_cas_sym <- pairs(emm_cas_sym, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "Caspase", state = "Sym")
letters_cas_sym <- cld(emm_cas_sym, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "Caspase", state = "Sym")

# Supplemental table
anova_table <- bind_rows(
```

```
  as.data.frame(anova_edu_apo)  %>% rownames_to_column("term") %>% mutate(marker = "EdU",     state = "
  as.data.frame(anova_edu_inoc) %>% rownames_to_column("term") %>% mutate(marker = "EdU",     state = "
  as.data.frame(anova_edu_sym)  %>% rownames_to_column("term") %>% mutate(marker = "EdU",     state = "
  as.data.frame(anova_cas_apo)  %>% rownames_to_column("term") %>% mutate(marker = "Caspase", state = "
  as.data.frame(anova_cas_inoc) %>% rownames_to_column("term") %>% mutate(marker = "Caspase", state = "
  as.data.frame(anova_cas_sym)  %>% rownames_to_column("term") %>% mutate(marker = "Caspase", state = "
) %>%
  select(marker, state, term, everything())

supplemental_tukey_table <- bind_rows(
  tukey_edu_apo,
  tukey_edu_inoc,
  tukey_edu_sym,
  tukey_cas_apo,
  tukey_cas_inoc,
  tukey_cas_sym
) %>%
  select(marker, state, time, contrast, estimate, SE, df, t.ratio, p.value) %>%
  mutate(
    estimate = round(estimate, 2),
    SE = round(SE, 2),
    t.ratio = round(t.ratio, 2),
    p.value = signif(p.value, 3)
  ) %>%
  arrange(marker, state, time)

write_csv(anova_global_table, "~/Documents/GitHub/heating_lacerates_final/tables/TableS10-confocal_anova
write_csv(supplemental_tukey_table, "~/Documents/GitHub/heating_lacerates_final/tables/TableS11-confocal
```